

CHANDRASHEKAR BALA

Cybersecurity Researcher | Offensive Security | VAPT & Security Engineering

✉ chandrashekar-bala@protonmail.com

☎ +91 9133093202

📍 Hyderabad, Telangana

🌐 in/chandrashekar-bala

🐙 github.com/chandrashekar-bala

🌐 chandrashekar-bala.github.io

SUMMARY

Cybersecurity professional focused on finding, validating, and understanding security weaknesses across applications, networks, systems, and security infrastructure. Hands-on work spans penetration testing, VAPT, web application security, network and wireless security, vulnerability research, exploit development, privilege escalation, post-exploitation analysis, malware analysis, digital forensics, threat intelligence, and security engineering. I approach security from both attacker and defender perspectives—reconstructing attack paths, validating exploitability, preserving and analyzing evidence, assessing impact, and translating findings into remediation, detection, and engineering improvements. Currently serving as Team Lead — Cybersecurity at NETWORKWALKS Technologies while continuing independent security research across offensive, defensive, adversary-research, and systems-engineering domains.

EXPERIENCE

Team Lead — Cybersecurity, NETWORKWALKS Technologies Oct'26 – Present
Security Assessment | Offensive Security | Defensive Security | Adversary Research | Security Engineering

- Lead and contribute to multidisciplinary cybersecurity work spanning security assessment, offensive security, defensive investigation, adversary research, security engineering, and technical project execution.
- Guide technical project work involving reconnaissance, vulnerability assessment, penetration testing, security analysis, investigation, evidence review, attack-path analysis, and security documentation.
- Coordinate technical work across assigned cybersecurity activities while maintaining an evidence-driven approach to validation, impact analysis, remediation, and reporting.
- Translate offensive findings and investigative observations into defensive considerations, engineering improvements, and practical security recommendations.

Cybersecurity Intern — NETWORKWALKS Technologies Sep'26 – Oct'26
VAPT | Network Security | Security Operations | Incident Response | Digital Forensics

- Worked across network security, vulnerability assessment, penetration testing, security operations, incident response, digital forensics, and risk assessment activities.
- Performed reconnaissance, vulnerability validation, technical analysis, investigation, evidence review, and security documentation across assigned cybersecurity work.
- Applied offensive-security and investigative techniques to identify weaknesses, validate findings, analyze attack paths and security events, assess impact, and document remediation considerations.

Cybersecurity Researcher — Independent Jul'24 – Present
Independent Cybersecurity Research & Security Engineering

- Conduct hands-on research across offensive security, vulnerability assessment, penetration testing, web application security, network security, wireless security, system security, threat intelligence, malware analysis, digital forensics, and security engineering.
- Perform reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation analysis, and attack-path reconstruction.
- Conduct web application security testing with Burp Suite and OWASP ZAP across SQL injection, XSS, authentication, authorization, access-control, session, and HTTP request/response behavior.
- Analyze OSINT sources, domains, IP addresses, infrastructure, threat reports, security indicators, and attacker TTPs; map relevant adversary behavior to MITRE ATT&CK.
- Analyze packet captures and network traffic using Wireshark to investigate protocols, communication patterns, suspicious connections, and attacker behavior.
- Perform Windows forensic and malware-analysis research involving registry artifacts, filesystem activity, disk evidence, suspicious files, execution behavior, and static/dynamic analysis.
- Develop Python and Bash tooling for reconnaissance, repeatable security analysis, proof-of-concept development, and security research workflows.
- Document findings, technical evidence, exploitation conditions, attack paths, security impact, remediation recommendations, and validation methodology.

FEATURED SECURITY WORK

Mediroza General Hospital — Authorized Black-Box Web Application Security Assessment Sep'26 – Oct'26
Web Application Security | Burp Suite | Nmap | Manual Validation | CVSS/CWE

- Conducted an authorized black-box assessment and reconstructed the attack path from reconnaissance through demonstrated SQL injection, authentication bypass, protected report retrieval, and impact analysis.
- Documented eight security findings spanning directory exposure, internal database-backup exposure, SQL injection, verbose database errors, sensitive report retrieval, predictable storage, error-log disclosure, and technology fingerprinting.
- Produced evidence-driven findings covering exploitation conditions, severity, CWE/CVSS context, root cause, security impact, remediation, and retest considerations.

RTL8812BU / RTL8822BU Linux Driver Modernization & Wireless Security Engineering Jul'26 – Aug'26
C | Linux Kernel | GCC | Git | Kali Linux | Real Hardware

- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using Linux kernel headers, GCC, conditional compilation, Git, and kernel build workflows.
- Built and validated the modified kernel module on real RTL8812BU hardware and exercised monitor mode, packet injection, packet capture, Wireshark, and Aircrack-ng workflows.

- Performed structured reconnaissance and service validation using WHOIS, DNS tooling, certificate discovery, web fingerprinting, Nmap/NSE, TLS inspection, and evidence-preserving scan artifacts.
- Correlated host, service, web, and infrastructure data to distinguish observed exposure from hypotheses and support subsequent security assessment.

Web Application Security Research
PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

Feb'26 – Present

- Investigated SQL injection, XSS, authentication, authorization, access-control, session, and HTTP behavior across 50+ PortSwigger Web Security Academy labs; reproduced findings and documented exploitation conditions and remediation considerations.

PROJECTS & TECHNICAL RESEARCH

Vulnerability Assessment & Privilege Escalation
VulnHub | Nmap | Linux | Metasploit

Security Research

- Perform reconnaissance, service enumeration, vulnerability assessment, system enumeration, controlled exploitation, Linux privilege escalation, and post-exploitation analysis across isolated virtual environments.
- Trace attack paths from initial access through elevated privileges and document exploitation conditions, evidence, security impact, and hardening recommendations.

Buffer Overflow Exploit Development
C | Linux | GDB | Memory Analysis

Security Research

- Analyze stack-based buffer-overflow vulnerabilities and vulnerable program behavior using GDB, stack layout, registers, memory analysis, control-flow investigation, and controlled proof-of-concept workflows.

Enterprise Network Penetration Testing
Nmap | Metasploit | Linux | Network Security

Security Research

- Conduct network reconnaissance, service enumeration, vulnerability assessment, controlled validation, and attack-path analysis against isolated enterprise-style environments.
- Investigate exposed services and security weaknesses, validate exploitation paths, and document technical observations and remediation considerations.

Malware Analysis & Windows Forensics Research
Static/Dynamic Analysis | Ghidra | Volatility | Wireshark

Security Research

- Conduct controlled malware-analysis and forensic research involving suspicious files, registry artifacts, filesystem activity, disk evidence, execution behavior, indicator extraction, and timeline-oriented investigation.
- Use static and dynamic analysis techniques to examine binary behavior and correlate host and network evidence during investigation workflows.

Security Automation & Proof-of-Concept Tooling
Python | Bash | Linux | Security Research

Independent Research

- Develop Python and Bash tooling for reconnaissance, repeatable analysis, security testing, proof-of-concept development, and research automation.

TECHNICAL EXPERTISE

Offensive Security	Reconnaissance, service enumeration, vulnerability assessment, penetration testing, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis
Web & Application Security	OWASP Top 10, SQL Injection, XSS, authentication, authorization, access control, HTTP/HTTPS, request/response analysis, session behavior, Burp Suite, OWASP ZAP
Network & Wireless	TCP/IP, DNS, DHCP, routing, subnetting, CIDR, network reconnaissance, packet capture, traffic analysis, Nmap, Wireshark, Aircrack-ng, monitor mode, packet injection
Threat Intelligence	OSINT, threat intelligence, threat-report analysis, infrastructure analysis, domains/IPs, security indicators, TTP analysis, adversary research, MITRE ATT&CK
Malware & Forensics	Malware analysis, static/dynamic analysis, Windows forensics, registry analysis, filesystem and disk analysis, suspicious-file investigation, memory analysis, Ghidra, Volatility
Exploit Research	Buffer-overflow analysis, C, Linux, GDB, stack and memory analysis, control-flow investigation, proof-of-concept development
Security Engineering	Linux security, kernel-facing C, Linux kernel compatibility, wireless driver engineering, GCC, Git, Python/Bash automation, technical troubleshooting
Security Operations	Security monitoring, log analysis, alert triage, IOC analysis, threat detection, suspicious-activity investigation, incident response, evidence correlation

TOOLS & TECHNICAL STACK

Web / Application	Burp Suite, OWASP ZAP, HTTP/HTTPS analysis, request/response analysis, authentication and access-control testing
Network / Discovery	Nmap, Wireshark, packet capture, service enumeration, network reconnaissance, protocol analysis, DNS tooling
Exploitation / Research	Metasploit, GDB, controlled exploitation, privilege escalation, post-exploitation, buffer-overflow research, proof-of-concept development
Malware / Forensics	Ghidra, Volatility, GDB, Wireshark, Windows forensic artifacts, registry analysis, filesystem/disk investigation
Threat Research	MITRE ATT&CK, OSINT, threat-report analysis, TTP analysis, infrastructure analysis, adversary research
Programming / Systems	Python, Bash, C, SQL, Linux, Kali Linux, Windows, GCC, GDB, Git, Linux kernel headers, kernel build workflows

SELECTED TECHNICAL EVIDENCE

- **8+ documented security findings:** evidence-driven findings from an authorized black-box web application security assessment, including severity, root cause, impact, remediation, and retest considerations.
- **13+ driver source files:** targeted Linux kernel compatibility work across a Realtek RTL8812BU/RTL8822BU driver, followed by build and real-hardware validation.
- **65+ evidence files preserved:** structured assessment evidence including source artifacts, screenshots, scan outputs, and technical validation material.
- **50+ PortSwigger labs:** hands-on web application security research covering SQL injection, authentication, access control, XSS, request/response behavior, and related attack paths.
- **Evidence-driven methodology:** observation → validation → investigation → impact → remediation → retest.

CERTIFICATIONS & PROFESSIONAL TRAINING

Google Cybersecurity Professional Certificate	Feb'26 <i>Coursera</i>
Security, networking, Linux, SQL, Python, SIEM workflows, threat detection, incident response, vulnerability management, risk assessment, access control, and security operations.	
Tata — Cybersecurity Analyst Job Simulation	Aug'26 <i>Forage</i>
IAM strategy assessment, solution design, implementation planning, and secure access workflows.	
Introduction to Blockchain and Web3	May'24 <i>edX</i>
Blockchain, decentralized systems, Web3 architecture, and security considerations.	

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering	2024
Lovely Professional University	Jalandhar, Punjab
Minor: Machine Learning	
Relevant study: Network Security, Ethical Hacking, Penetration Testing, Machine Learning, Automation	

ADDITIONAL TECHNICAL CAPABILITIES

VAPT Delivery	Reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis, technical reporting
Web Security Testing	HTTP analysis, SQL injection, XSS, authentication, authorization, access-control testing, request/response analysis, Burp Suite, OWASP ZAP
Network Analysis	Network reconnaissance, service enumeration, exposed-service analysis, packet capture, traffic analysis, protocol analysis, Nmap, Wireshark
Exploit Research	Buffer-overflow analysis, memory analysis, GDB, C, Linux, control-flow investigation, proof-of-concept development
Threat Research	Threat intelligence, OSINT, threat-report analysis, TTP analysis, adversary behavior, infrastructure analysis, MITRE ATT&CK
Malware / Forensics	Static/dynamic analysis, suspicious-file investigation, Windows registry artifacts, filesystem activity, disk analysis, memory investigation
Security Engineering	Linux security, wireless security, kernel compatibility, C systems programming, Python/Bash automation, Git, GCC, technical troubleshooting
Security Documentation	Technical evidence, vulnerability impact, exploitation conditions, attack paths, remediation recommendations, retesting, security reporting

ADDITIONAL TRAINING

- **Java Full Stack Developer Training** — Feb'23 | CipherSchools
- **Google Cloud Facilitator Program** — May'22 | Google